

and ensure the residual risk will never increase, regardless of the number of shares that may or may not transact. The free block order resolves anxieties traders may have about adverse selection. For example, if an investor is buying \$150K of stock RLK and selling \$100K of stock LMK (with correlation = 1) the investor has market exposure of \$50K due to the higher value in RLK. Hence, the investor could enter \$50K of RLK into a dark pool and will always be better off if any shares are executed. This \$50K represents the “free block order” or “free block cross.” This number of shares should always be entered into a dark pool to reduce market risk.

APPENDIX

The matrices for the QP technique in Equation 9.7 are calculated as follows:

$$\tilde{r}_{m \cdot (n+1) \times 1} = \begin{pmatrix} \begin{bmatrix} r_{11} \\ \vdots \\ r_{1,n+1} \end{bmatrix} \\ \begin{bmatrix} r_{21} \\ \vdots \\ r_{2,n+1} \end{bmatrix} \\ \vdots \\ \begin{bmatrix} r_{m1} \\ \vdots \\ r_{m,n+1} \end{bmatrix} \end{pmatrix}$$

$$Q_{m \cdot (n+1) \times m \cdot (n+1)} = \tilde{M}_{m \cdot (n+1) \times m \cdot (n+1)} + 2\lambda^* \cdot \tilde{C}_{m \cdot (n+1) \times m \cdot (n+1)},$$

where,

$$\tilde{M} = \begin{pmatrix} M_1 & 0 & 0 & \cdots & 0 \\ 0 & M_2 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & & \vdots \\ \vdots & \vdots & & M_{m-1} & 0 \\ 0 & 0 & & 0 & M_m \end{pmatrix}$$

$$M_i = \begin{pmatrix} I'_i & -I'_i & 0 & \cdots & 0 \\ -I'_i & 2I'_i & 0 & \cdots & 0 \\ 0 & 0 & \ddots & & \vdots \\ \vdots & \vdots & & 2I'_i & -I'_i \\ 0 & 0 & & -I'_i & I'_i \end{pmatrix}$$

$$I_i = \frac{n \cdot b_1 I_i}{X_i V_i}$$